

EA-0010 - AQELYN Engineering Archive

AQELYN - EA-0010 Engineering Archive

IS-010 - AQELYN Compliance & Governance Engine

****Archive ID:**** EA-0010

****Implementation Specification:**** IS-010

****Component:**** AQELYN Compliance & Governance Engine

****Project:**** AQELYN

****System Type:**** Cyber Security Operating Environment

****Status:**** COMPLETE

****Repository Impact:**** No top-level repository structure changes

****Breaking Changes:**** None

****Engineering Phase:**** Phase 3

****Predecessor Archives:**** EA-0001 through EA-0009

****Next Specification:**** IS-011

Document Control

| Field | Value |

|---|---|

| Document | Engineering Archive EA-0010 |

| Specification | IS-010 - AQELYN Compliance & Governance Engine |

| Source Basis | Rebuilt from the EA-0010 chat material and uploaded source PDF |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-010 complete; EA-0010 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment. The project has completed the first nine engineering archives:

| Engineering Archive | Implementation Specification | Component |

|---|---|---|

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

| EA-0004 | IS-004 | AQELYN Evidence Engine |

EA-0005	IS-005	AQELYN Knowledge Graph
EA-0006	IS-006	AQELYN Trust Engine
EA-0007	IS-007	AQELYN Mission Engine
EA-0008	IS-008	AQELYN Workflow Engine
EA-0009	IS-009	AQELYN Policy Engine
EA-0010	IS-010	AQELYN Compliance & Governance Engine

The fixed repository structure remains:

```
AQELYN/  
■■■ archive/  
■■■ blueprint/  
■■■ docs/  
■■■ src/  
■■■ tests/  
■■■ tools/  
■■■ build/  
■■■ releases/  
■■■ scripts/  
■■■ assets/  
■■■ examples/  
■■■ plugins/  
■■■ sdk/  
■■■ api/  
■■■ README.md
```

The engineering rule remains:

```
Finish Implementation Specification  
↓  
Generate Engineering Archive  
↓  
Continue
```

No Engineering Archive may be skipped.

2. IS-010 Specification Identity

```
Specification ID: IS-010  
Name: AQELYN Compliance & Governance Engine  
Engineering Archive Target: EA-0010  
Project: AQELYN  
System Type: Cyber Security Operating Environment  
Status: Complete  
Predecessor: IS-009 - AQELYN Policy Engine
```

3. Purpose

The AQELYN Compliance & Governance Engine provides AQELYN with a formal mechanism for managing cybersecurity compliance, governance obligations, control frameworks, audit readiness, accountability, and regulatory alignment.

It transforms AQELYN from a security automation platform into a governed cyber operating environment capable of answering:

```
Are we compliant?  
With what?  
Based on which controls?  
Supported by what evidence?  
Approved by whom?  
Auditable by whom?  
Changed when?
```

4. Core Mission

The engine shall:

- Map compliance frameworks
- Track governance obligations
- Evaluate control status
- Link controls to evidence
- Support audit readiness
- Integrate with policies, workflows, trust, knowledge, and evidence
- Maintain immutable governance traceability

5. Scope

5.1 In Scope

IS-010 includes:

- Compliance framework modeling
- Control catalog management
- Governance obligation tracking
- Control-to-policy mapping
- Control-to-evidence mapping
- Audit readiness state
- Governance decision records
- Regulatory alignment metadata
- Compliance event generation
- Compliance scoring
- Exception and waiver tracking
- Integration with existing AQELYN engines

5.2 Out of Scope

IS-010 does not implement:

- Full legal interpretation engine
- External regulator submission system
- Human legal advisory function
- Third-party GRC product replacement
- Financial compliance certification
- Automated legal liability decisions

6. Existing Engine Dependencies

The Compliance & Governance Engine depends on:

- IS-001 - AQELYN Kernel
- IS-002 - Universal Object Model
- IS-003 - AQELYN Event Bus
- IS-004 - AQELYN Evidence Engine
- IS-005 - AQELYN Knowledge Graph
- IS-006 - AQELYN Trust Engine
- IS-007 - AQELYN Mission Engine
- IS-008 - AQELYN Workflow Engine
- IS-009 - AQELYN Policy Engine

7. Architectural Role

The engine acts as the governance layer above policy enforcement.

Policy Engine:

Decides what is allowed, denied, required, or constrained.

Compliance & Governance Engine:

Explains why those rules exist, which controls they satisfy, what evidence supports them, and whether the organization is audit-ready.

8. High-Level Architecture

AQELYN Compliance & Governance Engine

- Framework Registry
- Control Catalog
- Obligation Manager
- Governance Decision Ledger
- Compliance Mapping Service
- Control Evaluation Service
- Evidence Binding Service
- Audit Readiness Service
- Exception & Waiver Manager
- Compliance Reporting Service
- Governance Event Publisher

9. Primary Concepts

9.1 Compliance Framework

A structured set of external or internal requirements.

Examples:

- ISO 27001
- NIST CSF
- SOC 2
- GDPR
- HIPAA
- PCI DSS
- Internal Security Baseline
- Internal Engineering Governance Standard

9.2 Control

A security or governance requirement that must be satisfied.

Example:

Control ID: ACCESS-001
Name: Privileged Access Review
Requirement: Privileged accounts must be reviewed periodically.
Evidence Required: Access review logs, approval records, identity inventory.

9.3 Obligation

A binding requirement derived from a framework, policy, contract, law, or internal governance rule.

9.4 Evidence Binding

A link between a compliance control and evidence collected by the AQELYN Evidence Engine.

9.5 Governance Decision

A recorded human or system decision that affects compliance posture.

Examples:

```
Control accepted
Control failed
Exception granted
Waiver approved
Risk accepted
Policy override approved
Audit finding closed
```

10. Functional Requirements

FR-010-001 - Framework Registry

The system shall maintain a registry of compliance frameworks.

Each framework shall include:

```
framework_id
name
version
authority
jurisdiction
effective_date
status
control_groups
metadata
```

FR-010-002 - Control Catalog

The system shall maintain a normalized catalog of controls.

Each control shall include:

```
control_id
framework_id
control_family
title
description
requirement_text
severity
implementation_status
evidence_requirements
linked_policies
linked_workflows
linked_assets
```

FR-010-003 - Obligation Tracking

The system shall track obligations created from:

```
Compliance frameworks
Internal policies
Risk decisions
Mission requirements
Regulatory requirements
Contractual requirements
Security baselines
```

FR-010-004 - Policy Mapping

The engine shall map compliance controls to AQELYN policies from IS-009.

Example:

```
Control: ACCESS-001
Mapped Policy: POL-IDENTITY-004
Policy Effect: Require MFA for privileged users
Compliance Purpose: Supports access control requirement
```

FR-010-005 - Evidence Mapping

The engine shall bind controls to evidence objects from IS-004.

Example:

```
Control: LOGGING-002
Evidence:
- authentication_logs
- event_bus_audit_records
- kernel_execution_records
```

FR-010-006 - Compliance Evaluation

The engine shall evaluate each control state as:

```
Compliant
Partially Compliant
Non-Compliant
Not Applicable
Unknown
Waived
Exception Approved
```

FR-010-007 - Governance Decisions

The system shall record governance decisions with:

```
decision_id
decision_type
actor
authority_level
reason
scope
affected_controls
timestamp
evidence_links
approval_chain
expiry_date
```

FR-010-008 - Exceptions and Waivers

The engine shall support controlled exceptions and waivers.

Each waiver shall include:

```
waiver_id
control_id
reason
risk_acceptance
approved_by
valid_from
valid_until
compensating_controls
review_required
```

FR-010-009 - Audit Readiness

The engine shall calculate audit readiness based on:

- Control status
- Evidence completeness
- Policy coverage
- Workflow completion
- Trust score
- Open exceptions
- Expired waivers
- Unresolved findings

FR-010-010 - Reporting

The engine shall generate reports for:

- Framework compliance
- Control gaps
- Evidence completeness
- Governance decisions
- Audit readiness
- Exception register
- Waiver register
- Policy-control mapping

11. Non-Functional Requirements

NFR-010-001 - Traceability

Every compliance decision shall be traceable to:

- Control
- Policy
- Evidence
- Actor
- Timestamp
- Workflow
- Governance decision

NFR-010-002 - Immutability

Governance decision records shall be append-only.

NFR-010-003 - Auditability

All compliance state changes shall produce audit events.

NFR-010-004 - Explainability

Every compliance status shall provide a reason.

Example:

Status: Partially Compliant
Reason: Required policy exists, but supporting evidence is incomplete.

NFR-010-005 - Security

Only authorized governance actors shall approve waivers, exceptions, or control status overrides.

NFR-010-006 - Interoperability

The engine shall expose normalized interfaces for:

Policies
Evidence
Knowledge graph nodes
Workflows
Trust scores
Mission requirements
Reports

12. Core Architecture Flow

```

Framework imported
↓
Controls normalized
↓
Controls mapped to policies
↓
Controls mapped to evidence requirements
↓
Evidence collected
↓
Control status evaluated
↓
Governance decisions recorded
↓
Audit readiness calculated
↓
Compliance report generated
↓
Compliance event published

```

13. Integration Map

```

Compliance Engine
■
■ ■ ■ AQELYN Kernel
■ ■ ■ ■ Runtime registration and lifecycle control
■
■ ■ ■ Universal Object Model
■ ■ ■ ■ Compliance objects, controls, obligations, decisions
■
■ ■ ■ Event Bus
■ ■ ■ ■ Publishes compliance.status.changed and governance.decision.recorded
■
■ ■ ■ Evidence Engine
■ ■ ■ ■ Provides evidence bindings and validation
■
■ ■ ■ Knowledge Graph
■ ■ ■ ■ Stores relationships between controls, policies, risks, assets, evidence
■
■ ■ ■ Trust Engine
■ ■ ■ ■ Provides trust scores for evidence, actors, and control confidence
■
■ ■ ■ Mission Engine
■ ■ ■ ■ Links compliance obligations to mission objectives
■
■ ■ ■ Workflow Engine
■ ■ ■ ■ Runs remediation, review, approval, and audit workflows
■
■ ■ ■ Policy Engine
■ ■ ■ ■ Maps policies to compliance controls

```

14. Internal Component Architecture

The Compliance & Governance Engine consists of independent services communicating through the AQELYN Event Bus.

```

Compliance & Governance Engine
■
■ ■ ■ Compliance Framework Registry

```

- Control Registry
- Control Evaluation Engine
- Governance Decision Ledger
- Obligation Manager
- Evidence Correlation Service
- Compliance Scoring Engine
- Audit Readiness Analyzer
- Exception & Waiver Manager
- Reporting Service
- Compliance API
- Event Publisher

Each component shall be independently testable and replaceable without affecting other AQELYN engines.

15. Component Specifications

15.1 Compliance Framework Registry

Responsibilities:

- Store framework definitions
- Version frameworks
- Register framework metadata
- Maintain framework lifecycle

Inputs:

- Framework packages
- Framework updates
- Administrator actions

Outputs:

- Framework objects
- Framework events
- Version records

15.2 Control Registry

Stores every normalized compliance control.

Each control contains:

- Control ID
- Framework ID
- Version
- Owner
- Classification
- Priority
- Status
- Evidence Requirements
- Linked Policies
- Linked Assets
- Linked Risks

15.3 Control Evaluation Engine

Responsible for evaluating compliance.

Evaluation sources include:

- Evidence Engine
- Policy Engine
- Trust Engine
- Workflow Engine

Knowledge Graph
Mission Engine

Possible results:

PASS
FAIL
PARTIAL
UNKNOWN
NOT_APPLICABLE
WAIVED

15.4 Governance Decision Ledger

Maintains immutable governance decisions.

Example:

```

Decision
↓
Approver
↓
Evidence
↓
Risk
↓
Approval Chain
↓
Timestamp
↓
Digital Signature
↓
Archive

```

This ledger shall be append-only.

15.5 Obligation Manager

Tracks obligations from:

Policies
Regulations
Frameworks
Contracts
Mission requirements
Internal standards

Each obligation shall support lifecycle management.

15.6 Evidence Correlation Service

Links evidence objects to compliance controls.

Example:

```

Evidence
↓
Control
↓
Requirement
↓
Evaluation
↓
Audit Package

```

The service consumes evidence without modifying the original evidence records.

15.7 Compliance Scoring Engine

Calculates overall compliance health.

Inputs:

- Control Status
- Trust Score
- Evidence Quality
- Policy Coverage
- Exceptions
- Waivers
- Audit Findings

Outputs:

- Framework Score
- Business Unit Score
- Mission Score
- Asset Score
- Organization Score

15.8 Audit Readiness Analyzer

Produces readiness indicators.

Example:

- Audit Ready
- Nearly Ready
- Missing Evidence
- High Risk
- Requires Review
- Failed

15.9 Exception & Waiver Manager

Maintains approved deviations.

Supports:

- Exception Requests
- Risk Acceptance
- Compensating Controls
- Waiver Expiration
- Automatic Review
- Renewal Workflow

15.10 Reporting Service

Produces:

- Executive Reports
- Auditor Reports
- Technical Reports
- Evidence Packages
- Compliance Dashboards
- Control Heat Maps
- Trend Analysis

16. Universal Object Model Extensions

The following objects extend IS-002.

16.1 ComplianceFramework

```
Framework:
framework_id
name
version
authority
jurisdiction
publication_date
effective_date
status
metadata
```

16.2 ComplianceControl

```
Control:
control_id
framework_id
family
title
description
severity
owner
status
evidence_requirements
```

16.3 GovernanceDecision

```
Decision:
decision_id
actor
authority
decision_type
reason
timestamp
signature
```

16.4 Obligation

```
Obligation:
obligation_id
source
owner
priority
due_date
status
```

16.5 Waiver

```
Waiver:
waiver_id
control
approver
reason
valid_from
valid_until
compensating_controls
```

16.6 AuditFinding

```
Finding:
finding_id
severity
affected_controls
recommendation
owner
remediation_status
```

17. Knowledge Graph Integration

The Knowledge Graph shall create relationships such as:

```

Framework
↓
contains
↓
Control
↓
implemented_by
↓
Policy
↓
verified_by
↓
Evidence
↓
owned_by
↓
Organization
↓
approved_by
↓
Governance Decision
  
```

Additional relationships include:

```

Control
↓
protects
↓
Asset
↓
supports
↓
Mission
↓
reduces
↓
Risk
  
```

18. Event Bus Integration

The engine publishes events defined by IS-003.

18.1 Framework Events

```

framework.registered
framework.updated
framework.deprecated
  
```

18.2 Control Events

```

control.created
control.modified
control.deleted
control.evaluated
  
```

18.3 Compliance Events

```

compliance.status.changed
compliance.score.updated
compliance.report.generated
  
```

18.4 Governance Events

```

governance.decision.recorded
governance.approval.completed
governance.exception.approved
governance.waiver.expired
  
```

18.5 Audit Events

```
audit.package.created  
audit.readiness.updated  
audit.finding.opened  
audit.finding.closed
```

19. Evidence Engine Integration

Consumes:

```
Evidence Objects  
Evidence Hashes  
Evidence Metadata  
Evidence Chains  
Digital Signatures
```

The Compliance & Governance Engine never modifies evidence. It only references immutable evidence identifiers.

20. Policy Engine Integration

Maps:

```
Policy  
↓  
implements  
↓  
Compliance Control
```

Multiple policies may satisfy one control. One policy may satisfy multiple controls.

21. Workflow Engine Integration

Workflow examples:

```
New Control  
↓  
Assign Owner  
↓  
Collect Evidence  
↓  
Evaluate  
↓  
Review  
↓  
Approve  
↓  
Archive
```

Exception workflow:

```
Request  
↓  
Risk Assessment  
↓  
Approval  
↓  
Temporary Waiver  
↓  
Expiration Review
```

22. Trust Engine Integration

Compliance confidence is weighted using trust metrics.

Example:

```
Evidence Trust
+
Policy Trust
+
Actor Trust
+
Workflow Integrity
↓
Compliance Confidence
```

The resulting confidence score complements, but does not replace, the compliance status.

23. Mission Engine Integration

Each mission may define mandatory controls.

Example:

```
Mission
↓
Required Controls
↓
Policy Verification
↓
Evidence Collection
↓
Mission Ready
```

Mission readiness calculations may incorporate compliance status where defined by mission policy.

24. Public API

24.1 Framework API

```
GET /frameworks
POST /frameworks
GET /frameworks/{id}
PUT /frameworks/{id}
```

24.2 Control API

```
GET /controls
POST /controls
GET /controls/{id}
PUT /controls/{id}
DELETE /controls/{id}
```

24.3 Compliance API

```
GET /compliance/status
GET /compliance/report
POST /compliance/evaluate
```

24.4 Governance API

```

POST /governance/decision
POST /governance/waiver
POST /governance/exception

```

24.5 Audit API

```

GET /audit/readiness
GET /audit/findings
GET /audit/package

```

25. Repository Impact

No repository restructuring is introduced.

Expected implementation locations:

```

AQELYN/
  ■■■ src/
  ■ ■■■ compliance/
  ■■■ tests/
  ■ ■■■ compliance/
  ■■■ docs/
  ■ ■■■ compliance/
  ■■■ api/
  ■ ■■■ compliance/
  ■■■ archive/

```

This conforms to the fixed repository standard established for AQELYN.

26. Security Architecture

The Compliance & Governance Engine is a high-trust subsystem. Its primary responsibility is to ensure that compliance state, governance decisions, and audit evidence remain authentic, traceable, and resistant to unauthorized modification.

26.1 Security Principles

The engine shall implement:

```

Zero Trust
Least Privilege
Defense in Depth
Immutable Audit Trail
Cryptographic Integrity
Separation of Duties
Need-to-Know Access
Complete Traceability

```

26.2 Authorization Model

Governance operations require authenticated and authorized actors.

Example roles:

Role	Capabilities
Compliance Administrator	Manage frameworks and controls
Auditor	Read reports and evidence
Governance Board	Approve waivers and exceptions

Security Officer	Review findings and remediation
System Administrator	Operate infrastructure only; no governance override by default
Automation Service	Execute approved workflows within assigned permissions

Role assignments shall be evaluated through the AQELYN Policy Engine.

26.3 Separation of Duties

Critical governance actions shall require separation between implementation and approval where configured.

Examples include:

```
Control approval
Risk acceptance
Waiver approval
Exception approval
Audit closure
```

The engine shall support configurable multi-party approval workflows.

26.4 Integrity Protection

Governance records shall be protected through integrity mechanisms.

Each governance record shall support:

```
Unique Identifier
Creation Timestamp
Originating Actor
Digital Signature, where configured
Evidence References
Version Identifier
Immutable History
```

Records shall be append-only. Corrections are recorded as new entries rather than modifications.

27. Governance Lifecycle

Every compliance control follows a defined lifecycle.

```
Draft
↓
Registered
↓
Assigned
↓
Implemented
↓
Evidence Collection
↓
Evaluation
↓
Approved
↓
Operational Monitoring
↓
Periodic Review
↓
Retired
```

27.1 Control Review

Controls shall support scheduled reviews.

Review inputs include:

Policy changes
Framework updates
New evidence
Risk changes
Mission changes
Audit findings

27.2 Exception Lifecycle

Requested
↓
Risk Assessment
↓
Approval Review
↓
Exception Granted
↓
Monitoring
↓
Expiration
↓
Renewal or Closure

Exceptions shall not become permanent by default.

27.3 Waiver Lifecycle

Requested
↓
Business Justification
↓
Risk Acceptance
↓
Compensating Controls
↓
Approval
↓
Expiration Review

Expired waivers shall generate governance events.

28. Compliance Evaluation Model

Each control shall be evaluated using multiple factors.

Example inputs:

Evidence Completeness
Policy Coverage
Trust Confidence
Workflow Completion
Manual Assessment
Known Exceptions
Known Waivers
Mission Constraints

Possible evaluation output:

PASS
FAIL
PARTIAL
UNKNOWN
NOT APPLICABLE
WAIVED
UNDER REVIEW

The engine shall retain the evaluation rationale alongside the result.

29. Audit Package Generation

The engine shall be capable of assembling an audit package without modifying source records.

An audit package may include:

- Framework Information
- Control Definitions
- Evidence References
- Policy References
- Governance Decisions
- Risk Acceptance Records
- Exception Register
- Waiver Register
- Audit Findings
- Evaluation Summary

Packages shall reference immutable evidence identifiers.

30. Reporting Model

Reports may be generated for different audiences.

30.1 Executive Report

Includes:

- Overall Compliance Status
- Top Risks
- Framework Coverage
- Strategic Trends

30.2 Operational Report

Includes:

- Control Status
- Evidence Gaps
- Remediation Progress
- Pending Reviews

30.3 Auditor Report

Includes:

- Evidence References
- Evaluation Results
- Decision History
- Exception Register

30.4 Engineering Report

Includes:

- Policy Mapping
- Control Relationships
- Workflow Status
- Knowledge Graph References

31. Failure Handling

The engine shall continue operating safely under degraded conditions.

31.1 Missing Evidence

Status:
PARTIAL

Reason:
Evidence unavailable

31.2 Missing Policy

Status:
FAIL

Reason:
No implementing policy exists

31.3 Trust Engine Unavailable

Status:
UNDER REVIEW

Reason:
Confidence calculation unavailable

31.4 Event Bus Delay

Evaluations shall be queued and processed once event delivery resumes.

32. Performance Requirements

The engine shall support:

Incremental control evaluation
Event-driven updates
Cached framework metadata
Asynchronous report generation
Parallel evidence correlation where appropriate

Long-running report generation shall not block governance operations.

33. Scalability Requirements

The architecture shall support:

Multiple organizations, tenant-aware deployments
Thousands of frameworks
Hundreds of thousands of controls
Millions of evidence references
Large governance decision histories

Scalability mechanisms are implementation-dependent but shall preserve functional behavior.

34. Testing Strategy

34.1 Unit Testing

Validate:

- Framework registry
- Control evaluation
- Scoring logic
- Exception handling
- Waiver lifecycle
- Decision ledger
- Reporting components

34.2 Integration Testing

Verify interaction with:

- AQELYN Kernel
- Event Bus
- Policy Engine
- Evidence Engine
- Knowledge Graph
- Trust Engine
- Workflow Engine
- Mission Engine

34.3 System Testing

Validate complete scenarios including:

- Framework import
- Control evaluation
- Evidence binding
- Governance approval
- Audit package generation
- Compliance reporting

34.4 Security Testing

Verify:

- Authorization enforcement
- Separation of duties
- Integrity protections
- Immutable governance records
- Audit trail completeness

34.5 Regression Testing

Ensure existing engines, IS-001 through IS-009, continue to operate without behavioral changes introduced by IS-010.

35. Acceptance Criteria

IS-010 shall be considered complete when:

- Compliance frameworks can be registered and versioned.
- Controls can be modeled and managed.
- Controls can be mapped to AQELYN policies.
- Controls can reference immutable evidence.
- Governance decisions are recorded with traceability.
- Exceptions and waivers are lifecycle-managed.
- Audit readiness can be calculated.
- Compliance reports can be generated.
- Integration points with IS-001 through IS-009 are defined.
- Repository structure remains unchanged.

Testing strategy is documented.

36. Repository Validation

This specification introduces no changes to the agreed repository layout.

```
AQELYN/
■■■ archive/
■■■ blueprint/
■■■ docs/
■■■ src/
■■■ tests/
■■■ tools/
■■■ build/
■■■ releases/
■■■ scripts/
■■■ assets/
■■■ examples/
■■■ plugins/
■■■ sdk/
■■■ api/
■■■ README.md
```

Implementation is expected within the existing directories, such as `src/compliance`, `tests/compliance`, and `api/compliance`, without altering the top-level structure.

37. Completion Summary

IS-010 - AQELYN Compliance & Governance Engine defines a governance subsystem that:

```
Models compliance frameworks and controls.
Connects controls to policies, evidence, missions, and governance decisions.
Maintains immutable governance records.
Supports exception and waiver management.
Calculates audit readiness.
Produces compliance reporting.
Integrates with all previously completed AQELYN engines.
Preserves the fixed repository structure and engineering methodology.
```

38. Specification Status

```
Specification ID : IS-010
Title : AQELYN Compliance & Governance Engine
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0010
```

Engineering workflow status:

```
EA-0001 COMPLETE
EA-0002 COMPLETE
EA-0003 COMPLETE
EA-0004 COMPLETE
EA-0005 COMPLETE
EA-0006 COMPLETE
EA-0007 COMPLETE
EA-0008 COMPLETE
EA-0009 COMPLETE
IS-010 COMPLETE
EA-0010 READY TO GENERATE
```

The implementation specification is complete. According to the AQELYN engineering rules, the next artifact must be EA-0010, after which work can continue with IS-011.

39. EA-0010 Engineering Objective

The objective of IS-010 was to introduce a dedicated Compliance & Governance Engine that enables AQELYN to manage regulatory compliance, governance obligations, audit readiness, and decision traceability as first-class capabilities within the Cyber Security Operating Environment.

The engine extends AQELYN beyond policy enforcement by providing structured governance, evidence-backed compliance evaluation, and immutable governance records.

40. EA-0010 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

- Compliance framework registration
- Control catalog management
- Governance obligation tracking
- Policy-to-control mapping
- Evidence-to-control correlation
- Compliance evaluation
- Governance decision recording
- Exception and waiver lifecycle management
- Audit readiness analysis
- Compliance reporting

The engine integrates with all previously completed AQELYN engines while preserving architectural modularity.

41. Major Engineering Decisions

41.1 Decision 1 - Dedicated Governance Layer

Compliance responsibilities are implemented as a standalone engine rather than extending the Policy Engine.

Rationale:

- Clear separation of concerns.
- Independent lifecycle.
- Improved maintainability.
- Better support for regulatory frameworks.

41.2 Decision 2 - Immutable Governance Ledger

Governance decisions are append-only.

Benefits:

- Full audit traceability.
- Historical accountability.
- Non-repudiation support.
- Consistent forensic evidence.

41.3 Decision 3 - Evidence Referencing

Compliance records reference immutable evidence maintained by the AQELYN Evidence Engine.

Benefits:

```
Single source of truth.
No duplication of evidence.
Consistent audit chains.
Reduced storage overhead.
```

41.4 Decision 4 - Event-Driven Architecture

Compliance state changes are published through the AQELYN Event Bus.

Examples include:

```
compliance.status.changed
governance.decision.recorded
audit.readiness.updated
framework.registered
```

This maintains loose coupling between AQELYN engines.

41.5 Decision 5 - Universal Object Model Extension

New domain objects introduced include:

```
ComplianceFramework
ComplianceControl
GovernanceDecision
Obligation
Waiver
AuditFinding
```

These extend the Universal Object Model without modifying existing object definitions.

42. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle |

| IS-002 Universal Object Model | Governance objects |

| IS-003 Event Bus | Compliance events |

| IS-004 Evidence Engine | Evidence binding |

| IS-005 Knowledge Graph | Relationship storage |

| IS-006 Trust Engine | Confidence weighting |

| IS-007 Mission Engine | Mission compliance |

| IS-008 Workflow Engine | Review and approval workflows |

| IS-009 Policy Engine | Policy-to-control mapping |

No existing engine required redesign.

43. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/
■■■ src/compliance/
■■■ tests/compliance/
■■■ api/compliance/
■■■ docs/compliance/
■■■ archive/
```

No top-level directories were added, removed, or renamed.

44. Security Impact Summary

The specification introduces governance-specific security controls:

```
Role-based authorization
Separation of duties
Immutable governance records
Traceable approval chains
Cryptographic integrity support
Audit-ready decision history
```

No reduction in the security posture of existing components was identified.

45. Compliance Capabilities Added

The engine enables support for governance across frameworks such as:

```
ISO/IEC 27001
NIST Cybersecurity Framework
SOC 2
GDPR
HIPAA
PCI DSS
Internal security baselines
Organization-specific governance standards
```

The architecture is framework-agnostic, allowing additional frameworks to be incorporated without redesign.

46. Risks Identified

| Risk | Mitigation |

|---|---|

| Large framework datasets | Modular registries and scalable storage |

| Evidence incompleteness | Explicit evaluation states and audit readiness indicators |

| Unauthorized governance actions | RBAC and separation of duties |

| Framework evolution | Versioned framework registry |

| Long-running reports | Asynchronous report generation |

No critical architectural risks were identified that require redesign.

47. Verification Summary

The specification defines verification for:

```

Unit testing
Integration testing
System testing
Security testing
Regression testing

```

Acceptance criteria cover framework management, control evaluation, evidence mapping, governance decisions, reporting, and integration with all existing AQELYN engines.

48. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

```

Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility

```

49. Dependencies

Required:

```

EA-0001 through EA-0009
IS-001 through IS-009

```

Enables:

```

IS-011 and subsequent governance-dependent components

```

50. Completion Record

```

Engineering Archive : EA-0010
Implementation Specification : IS-010
Title : AQELYN Compliance & Governance Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
Engineering Rule :
IS Completed
↓
EA Generated
↓
Continue

```

51. Archive Index Update

```

EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine

```

52. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0010

Current Status:
EA-0010 COMPLETE

Next Implementation Specification:
IS-011 - AQELYN Identity & Access Governance Engine

EA-0010 is completed and archived. The engineering workflow is consistent with the project rule:

Implementation Specification -> Engineering Archive -> Continue

From this point onward, the next engineering artifact is IS-011.

53. Engineering Archive Publication Standard

From EA-0010 onward, every completed Engineering Archive shall be generated as a complete engineering package.

Required package structure:

```
EA-xxxx/
■
■ ■ ■ ■ diagrams/
■ ■ ■ ■ Architecture.svg
■ ■ ■ ■ Component.svg
■ ■ ■ ■ Workflow.svg
■ ■ ■ ■ EventFlow.svg
■ ■ ■ ■ Integration.svg
■
■ ■ ■ ■ examples/
■ ■ ■ ■ example_*.md
■
■ ■ ■ ■ HTML/
■ ■ ■ ■ index.html
■ ■ ■ ■ styles.css
■ ■ ■ ■ assets/
■
■ ■ ■ ■ images/
■
■ ■ ■ ■ journal/
■ ■ ■ ■ Engineering_Journal.md
■
■ ■ ■ ■ manifest/
■ ■ ■ ■ manifest.json
■
■ ■ ■ ■ MD/
■ ■ ■ ■ EA-xxxx.md
■
■ ■ ■ ■ PDF/
■ ■ ■ ■ EA-xxxx.pdf
■
■ ■ ■ ■ requirements/
■ ■ ■ ■ Requirements_Matrix.md
■
■ ■ ■ ■ traceability/
■ ■ ■ ■ Traceability_Matrix.md
■
■ ■ ■ ■ README.md
```

Each archive shall include:

```
Markdown engineering document
PDF documentation
HTML documentation with index.html
Manifest with metadata
Requirements matrix
Traceability matrix
```

Engineering journal
 Architecture/workflow diagrams when applicable
 Examples
 README
 ZIP package containing everything

The master Markdown is the source of truth. The PDF and HTML must be generated from the same master Markdown and must not omit sections.

54. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
FR-010-001	Maintain registry of compliance frameworks	Sections 10, 15	Complete
FR-010-002	Maintain normalized control catalog	Sections 10, 15	Complete
FR-010-003	Track governance obligations	Sections 10, 15	Complete
FR-010-004	Map controls to policies	Sections 10, 20	Complete
FR-010-005	Bind controls to evidence	Sections 10, 19	Complete
FR-010-006	Evaluate compliance status	Sections 10, 28	Complete
FR-010-007	Record governance decisions	Sections 10, 15, 26	Complete
FR-010-008	Support exceptions and waivers	Sections 10, 27	Complete
FR-010-009	Calculate audit readiness	Sections 10, 29, 30	Complete
FR-010-010	Generate compliance reports	Sections 10, 30	Complete
NFR-010-001	Traceability	Sections 11, 17, 50	Complete
NFR-010-002	Immutability	Sections 11, 26, 41	Complete
NFR-010-003	Auditability	Sections 11, 18, 29	Complete
NFR-010-004	Explainability	Sections 11, 28	Complete
NFR-010-005	Security	Sections 11, 26, 44	Complete
NFR-010-006	Interoperability	Sections 11, 13, 42	Complete

55. Traceability Matrix

Source	Target	Relationship
IS-010 Purpose	EA-0010 Objective	Defines why the engine exists
Framework Registry	FR-010-001	Implements framework lifecycle
Control Registry	FR-010-002	Implements normalized controls
Obligation Manager	FR-010-003	Implements obligation tracking
Policy Engine Integration	FR-010-004	Provides policy-to-control mapping
Evidence Engine Integration	FR-010-005	Provides evidence-to-control mapping
Control Evaluation Engine	FR-010-006	Produces compliance result
Governance Decision Ledger	FR-010-007	Records immutable governance decision
Exception & Waiver Manager	FR-010-008	Manages approved deviations
Audit Readiness Analyzer	FR-010-009	Calculates audit readiness
Reporting Service	FR-010-010	Generates compliance and governance reports
Security Architecture	NFR-010-005	Enforces authorization and integrity
Event Bus Integration	NFR-010-003	Produces audit and compliance events

| Knowledge Graph Integration | NFR-010-001 | Enables relationship traceability |

| Repository Validation | Repository Standard | Confirms no top-level redesign |

56. Engineering Journal

Journal Entry - EA-0010

EA-0010 was created to archive the completion of IS-010 - AQELYN Compliance & Governance Engine.

The archive records the transition of AQELYN from policy enforcement into governed cyber operations. IS-010 defines the structure needed to model compliance frameworks, evaluate controls, bind controls to evidence, record governance decisions, manage waivers and exceptions, calculate audit readiness, and generate compliance reports.

The archive also establishes the Engineering Archive Publication Standard, requiring future EA packages to contain Markdown, PDF, HTML, manifest, requirements, traceability, journal, diagrams, examples, and README files.

Lessons Learned

The Engineering Archive must be generated from one master Markdown source. PDF and HTML versions must be rendered from that exact source to avoid summary artifacts and content loss.

Governance Note

EA-0010 is accepted as the first archive to formalize the package publication standard. Future archives shall follow the same structure by default.

57. Examples

57.1 Example Control Mapping

```
control_id: ACCESS-001
framework: Internal Security Baseline
title: Privileged Access Review
mapped_policy: POL-IDENTITY-004
evidence_required:
- privileged_user_inventory
- access_review_approval_log
- identity_provider_audit_log
status: PARTIAL
reason: Policy exists, but current review evidence is incomplete.
```

57.2 Example Governance Decision

```
decision_id: GOV-DEC-0001
decision_type: waiver_approved
actor: governance_board
authority_level: governance
affected_controls:
- ACCESS-001
reason: Temporary operational constraint with compensating controls.
evidence_links:
- evidence://access-review-2026-q1
valid_until: 2026-09-30
```

57.3 Example Compliance Event

```
{
  "event_type": "compliance.status.changed",
```

```
"control_id": "ACCESS-001",  
"previous_status": "UNKNOWN",  
"new_status": "PARTIAL",  
"reason": "Required evidence incomplete",  
"source_engine": "aqelyn_compliance_governance_engine"  
}
```

58. Manifest Summary

Archive contents include:

```
README.md  
MD/EA-0010.md  
PDF/EA-0010.pdf  
HTML/index.html  
HTML/styles.css  
manifest/manifest.json  
requirements/Requirements_Matrix.md  
traceability/Traceability_Matrix.md  
journal/Engineering_Journal.md  
diagrams/Architecture.svg  
diagrams/Component.svg  
diagrams/Workflow.svg  
diagrams/EventFlow.svg  
diagrams/Integration.svg  
examples/example_control_mapping.md  
source/EA-0010-copied-Allchat.pdf, when available
```

59. Final Archive Statement

EA-0010 is the Engineering Archive for IS-010 - AQELYN Compliance & Governance Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the publication standard for future AQELYN Engineering Archives.

```
EA-0010 COMPLETE  
IS-010 COMPLETE  
NEXT: IS-011
```